

PRODUCT REQUIREMENT DOCUMENT

StreakForge

Fitness Engagement & Retention Analytics Platform for an Indian Gym Chain

Document owner	Diya — Analytics / Data Product Lead
Status	Implemented — v1 delivered (Notebooks 01–06 + Streamlit app, PR1–PR21)
Document version	1.0
Last updated	17 August 2026
Pipeline cutoff (AS_OF_DATE)	2026-08-14 (see §4 for the data-realistic boundary duality)

1. Overview

Project name: StreakForge

Type: Data analytics / insight-generation platform (no predictive modelling in v1)

Output: A documented, end-to-end analytical workflow (Jupyter notebooks → processed CSV artefacts → Streamlit dashboard) that identifies which member engagement behaviours influence long-term retention for a multi-branch Indian gym chain.

This PRD documents StreakForge against the same standard used for any analytics product review: a specific, quantified business problem; named stakeholders; verified data sources; measurable KPIs; scoped user stories; a documented data workflow; and an explicit risk register. Because the v1 build is complete, sections below report both the planned requirement and the delivered outcome side by side, so the document remains a true record of what was built and why.

2. Business Problem Statement

2.1 Problem

Branch managers and the retention/growth team at the gym chain cannot currently answer a basic question: which member behaviours in the first weeks of membership predict whether a member stays or churns. Check-in logs, app engagement events, streak history, and renewal records are captured across four separate operational systems and never combined, so no one can see a member's full engagement picture in one place. Without that picture, retention spend (win-back calls, PT trial offers, onboarding nudges) is allocated on instinct rather than evidence.

2.2 Goal

Consolidate five independently sourced datasets (~309,500 rows total) into one validated analytical fact table, and surface the specific, evidence-backed levers that most affect activation and retention — so that Product, Growth, and Customer Success teams can act on a short list of confirmed findings rather than a wall of raw logs.

2.3 Primary users

Retention/Growth analysts and the Head of Member Experience at the gym chain's central analytics function, who currently rely on manual, per-branch spreadsheet pulls.

2.4 Success criteria (verifiable)

- A single merged, validated analytical dataset exists, joined on member_id with documented grain discipline (deduplication before every renewal- and streak-episode-level aggregation).
- At least 5 confirmed, evidence-backed insights are documented with business impact, audience, and recommended action (delivered: 5 — see §7).
- A live Streamlit dashboard is deployed and accessible without any analytical logic being re-computed inside the app (delivered — see §9).

- The single highest-leverage retention lever in the dataset is identified and named explicitly, not left as a vague 'engagement matters' conclusion (delivered — 7-day activation, not renewal, is the primary loss point; see §7, INS-02).

✓ **Specific · Quantified · Named users · Measurable success** This statement follows the PRD Playbook's three requirements: specificity, quantification, and verifiable success criteria.

3. Stakeholder Identification & Business Impact

3.1 Stakeholder map

Type	Who	Role in the product
Primary users	Retention / Growth analysts (central function)	Use the dashboard's KPI cards, at-risk member list, and funnel view daily/weekly to prioritise outreach and onboarding fixes.
Secondary users	Branch managers & Head of Member Experience	Review branch-level churn/activation comparisons periodically; consume exported insight summaries and executive one-pagers rather than the live app.
Data owners	Gym operations (check-in & branch systems), App engineering (event logs), Billing/finance (renewals)	Own the four source systems the pipeline reads from. Any schema or field-meaning change here must be re-verified against the data dictionary before the pipeline is re-run.
Approvers	Head of Member Experience (business sign-off); Analytics Lead (technical/methodology sign-off)	Sign off on the AS_OF_DATE duality framing (§4.3) and on which insights are cleared for leadership circulation before every reporting cycle.

3.2 Business impact justification

Operational impact: Engagement data is currently reconciled manually per branch; no standing pipeline exists to combine check-ins, app events, streaks, and renewals into one place, so root-cause questions ("why did this member churn?") take days rather than minutes to answer.

Revenue/cost impact: Median revenue per member is ₹8,970 and mean revenue per member is ₹18,711 (right-skewed by higher-tier plans); a churn rate materially above what onboarding-stage fixes could prevent represents recoverable revenue that is currently unquantified and unmanaged.

User experience impact: 12 currently-active members were identified as triggering nearly every composite risk signal simultaneously — a small, high-value, high-risk cohort (paying ₹48K–₹102K per logged session) that had no mechanism for being surfaced before this platform existed.

4. Dataset & Data Source Documentation

Every field referenced by the pipeline has been profiled against the actual CSV files (real column names, dtypes, and null rates) before any feature or KPI was built on top of it — no field is assumed. Full column-level detail is maintained in `data_dictionary_draft.csv` (53 columns profiled).

4.1 Source datasets

Dataset	Grain	Rows	Owner	Verified quality notes
members_master	One row per member	~17,500	Gym operations / CRM	date_of_birth 3.1% missing; all other fields complete.
subscription_renewal_records	One row per billing/renewal cycle	~46,000	Billing / finance	No missing values; drop_duplicates(renewal_id) enforced before any renewal-grain aggregation.
streak_history_episodes	One row per streak episode	~40,000	App engineering	streak_end_date & break_reason 3.1% missing (open/ongoing streaks); overlapping streak windows flagged, not silently fixed.
gym_checkin_workout_logs	One row per physical check-in	~108,000	Branch / access-control systems	checkout_datetime 12.1% missing; trainer_id 88.4% missing (most sessions are unassisted — expected, not an error).
app_engagement_events	One row per digital app event	~98,000	App engineering	notification_engagement 74.5% missing (only populated for notification-type events — expected by event_type).

`gym_checkin_workout_logs` and `app_engagement_events` are unioned into a single `engagement_events` fact table (~206,000 rows), then enriched via range/as-of joins against `members_master`, `subscription_renewal_records`, and `streak_history_episodes` on `member_id` plus the relevant date range.

4.2 Refresh & persistence

Refresh cadence: Batch — fresh data reload per notebook/PR section; no in-memory state is carried between pipeline stages, so every stage is independently reproducible.

Persistence format: CSV-only throughout the entire pipeline (data/processed/ and reports/) — a non-negotiable project constraint; no parquet, pickle, or other binary formats are used at any stage.

4.3 Known data artefact — AS_OF_DATE duality (documented, not an error)

The pipeline's fixed cutoff, AS_OF_DATE = 2026-08-14, sits well past the dataset's actual last recorded event (2025-10-05). Read literally, this inflates headline churn to 94.0%. Re-anchoring to the data-realistic boundary drops churn to 84.0% — still high, but the operationally meaningful figure. Both numbers are surfaced side by side on every relevant page and export; this is an explicit, flagged design decision, never silently corrected.

⚠ Golden rule applied *No feature or KPI in this platform was planned around a field whose type, null rate, or meaning had not been personally confirmed against the real CSVs first.*

5. KPI & Success Metric Planning

Every KPI below carries all four required elements: metric, measurement method, numeric target/current value, and reporting timeline.

KPI	Measurement method	Current value	Target / review cadence
Overall churn rate (headline)	Members with no engagement event after AS_OF_DATE – 90d, computed at pipeline cutoff (2026-08-14)	94.0%	Always reported alongside the data-realistic figure below; reviewed every dashboard refresh.
Churn rate (data-realistic)	Same method, re-anchored to last observed event date (2025-10-05)	84.0%	Treated as the operational KPI; target: reduce by tracking activation fixes below, reviewed monthly.
7-day activation rate	% of members with ≥ 1 logged engagement event within 7 days of join_date	18.8%	Sustain the observed upward cohort trend (14.6% → 36.6% by Sep-2025 cohort); reviewed monthly per signup cohort.
Renewal follow-through rate	% of eligible renewals completed vs. lapsed, at renewal-episode grain (deduplicated)	42.3%	Monitor for drop below 35% as an early-warning trigger; reviewed each billing cycle.
Streak-break recovery rate	% of broken streaks followed by a new streak episode within the product-defined recovery window	14.8%	Track quarter-over-quarter; no formal target set pending onboarding-fix rollout (see INS-02).

KPI	Measurement method	Current value	Target / review cadence
Avg weekly session rate	Mean engagement_events per member per week, member-deduplicated	0.30/wk	Directional indicator only; reviewed alongside activation cohort trend, monthly.
At-risk member list size	Count of active members triggering ≥ 3 composite risk signals simultaneously	12 members	Reviewed and re-run every billing cycle; each member routed to Customer Success outreach within that cycle.

6. User Stories & Stakeholder Workflow

Every story follows Role + Action + Business Benefit and is traceable to a delivered dashboard page or export.

- US-01: As a Retention/Growth analyst, I want to see the 7-day activation funnel with its drop-off stage highlighted, so that I can prioritise onboarding-flow experiments over undirected acquisition spend.
- US-02: As a Retention/Growth analyst, I want a pre-computed list of at-risk active members with their composite risk signals, so that I can hand Customer Success a short, actionable outreach list instead of scanning raw logs.
- US-03: As a branch manager, I want to compare my branch's churn and activation figures against the platform-wide distribution, so that I know whether an issue is local to my branch or systemic before requesting an operational review.
- US-04: As the Head of Member Experience, I want an executive one-pager that states both the headline and data-realistic churn figures side by side, so that I never present or act on a partially artefactual statistic.
- US-05: As a Retention/Growth analyst, I want to export the current filtered dashboard view as Markdown, a CSV bundle, or an emailed report, so that findings reach stakeholders who don't use the live app.
- US-06: As the Analytics Lead, I want every notebook and app page to flag data-quality issues (nulls, duplicates, overlapping streak windows) rather than silently fix them, so that downstream consumers know exactly what was and wasn't corrected.

✓ **Named role · Specific action · Explicit business benefit** All six stories pass the "so that" test — each has a justified reason to exist.

7. Feature Planning & Product Scope

7.1 In scope — v1 (delivered)

- ✓ Data acquisition verification, profiling, cleaning, and multi-source merging (Notebooks 01–02)
- ✓ Feature engineering for retention-relevant derived metrics: recency_days, plan_to_engagement_ratio, engagement_trend_slope, streak_break_recovery_flag, tenure_days_at_event (Notebook 03)
- ✓ Exploratory and behavioural analysis, incl. Cramér's V association tests and IQR-based outlier flagging (Notebook 04)
- ✓ KPI definition, business-metric design, and root-cause investigation workflow (Notebook 05)

- ✓ Visualisation and narrative-driven insight reporting using Plotly exclusively (Notebook 06)
- ✓ Streamlit dashboard: modular pages, filter/session-state system, light/dark theme, per-page error boundaries (PR17–PR18)
- ✓ Share & export: Markdown, CSV ZIP, optional PDF, SMTP email (PR19)
- ✓ Deployment configuration and a pre-flight health-check page (PR20–PR21)

7.2 Out of scope — v1

✗ Predictive / ML-based churn modelling ✗ Real-time streaming ingestion ✗ Production-grade multi-tenant authentication ✗ Re-deriving KPI or segment logic inside the Streamlit app (all analytical artefacts are pre-computed by the notebooks and only displayed by the app) ✗ Any persistence format other than CSV

 **Scope discipline applied** "App is read/display only" was enforced throughout: no KPI logic or segment definitions are re-derived inside the Streamlit app.

8. Data Workflow, Dashboard Planning & Risk Analysis

8.1 Data flow architecture

- 1. Ingestion:** Five raw CSVs read fresh per notebook/PR section — members_master, subscription_renewal_records, streak_history_episodes, gym_checkin_workout_logs, app_engagement_events.
- 2. Cleaning:** Explicit date re-parsing on every load; duplicates and overlapping streak windows flagged and annotated, not silently corrected; mixed date formats normalised.
- 3. Merging & feature engineering:** gym_checkin_workout_logs and app_engagement_events unioned into engagement_events (~206,000 rows), then range/as-of joined to members_master, subscription_renewal_records, and streak_history_episodes on member_id; grain discipline enforced via drop_duplicates(<business_key>) before every renewal- or streak-episode aggregation.
- 4. Analysis & KPI generation:** EDA, segmentation, RFM, and KPI computation write pre-aggregated outputs to data/processed/ and reports/figures/ (CSV only).
- 5. Visualisation & delivery:** Streamlit app reads only pre-computed artefacts via st.cache_data loaders with mtime-aware invalidation; renders KPI cards, funnel, at-risk list, branch comparisons, and executive one-pager; deployed on Streamlit Community Cloud with pinned requirements.txt and runtime.txt.

8.2 Risk & assumption register

Risk	Likelihood	Impact	Mitigation
Headline churn (94%) misquoted to leadership without the AS_OF_DATE context	Medium	High — risks panic win-back spend against a partly artefactual stat	Both churn figures (94.0% / 84.0%) shown side by side on every relevant page and export; executive one-pager states the duality explicitly.
Activation drop-off (81.2% at day 7) misread as a	Medium	High — would misdirect fixes	Root-cause investigation (INV-01) explicitly tested all segment columns; all Cramér's V < 0.1

Risk	Likelihood	Impact	Mitigation
single-cause, single-segment problem		toward channel/demographic targeting that the data rules out	documented before any recommendation was made.
A handful of branches wrongly blamed for platform-wide churn/activation issues	Low–Medium	Medium — would fund an unnecessary branch-level operational audit	Branch anomaly + activation compounding check (INV-03) run explicitly: 0 of 40 branches flagged; documented before recommending a centrally-focused fix instead.
Grain fan-out inflates renewal- or streak-level KPIs (double-counted members)	Medium	High — silently wrong KPI values with no visible symptom	drop_duplicates(<business_key>) enforced before every renewal- and streak-episode-grain aggregation, per project convention.
Stale cached dashboard data shown after a processed-file refresh	Low	Medium — analyst acts on outdated KPI values	st.cache_data loaders use mtime-aware cache invalidation; a pre-flight health-check page verifies freshness before use.
12-member at-risk list becomes stale as billing cycles roll over	Medium	Medium — Customer Success outreach targets outdated members	List is explicitly scoped as "re-run every billing cycle" rather than a one-time export (see §5 KPI table).

9. Dashboard & Delivery Plan

The Streamlit app is organised as read/display-only pages backed entirely by pre-computed CSV artefacts:

- Overview — KPI card grid (churn duality, activation, revenue, streak health) with global branch/date filters held in session state.
- Funnel & Activation — 7-day activation funnel, cohort trend chart, segment comparison (Cramér's V results surfaced, not hidden).
- Retention & Risk — at-risk member signal mix, branch churn/activation comparison, streak-break recovery view.
- Executive One-Pager — leadership-ready summary stating both churn figures and the top confirmed insights, exportable as Markdown / CSV ZIP / optional PDF / email.
- Health Check — pre-flight page confirming processed-file freshness, cache state, and pipeline version before an analyst trusts a number on screen.

Access: Deployed on Streamlit Community Cloud; theme toggle (light/dark) for extended daily use by analysts.

10. PRD Validation Checklist

- ✓ Is the business problem specific and quantified with evidence?
- ✓ Does every KPI have a numeric value/target, method, and review cadence?
- ✓ Are all stakeholders named with roles and ownership?
- ✓ Is the primary user clearly defined with a specific use case?
- ✓ Has the dataset been confirmed against the real CSVs (columns, dtypes, null rates)?
- ✓ Does every user story follow Role + Action + Business Benefit?
- ✓ Is v1 scope explicitly bounded with an out-of-scope list?
- ✓ Is the data workflow architecture documented end-to-end?
- ✓ Are all assumptions documented as identified risks with likelihood, impact, and mitigation?
- ✓ Is a dashboard/delivery plan included?
- ✓ Is the PRD free of aspirational language ("should", "might", "probably")?
- ✓ Are naming conventions (AS_OF_DATE, member_id, engagement_events, etc.) consistent throughout?
- ✓ Can a non-technical reader (e.g. Head of Member Experience) understand the document's purpose?
- ✓ Is the CSV-only persistence constraint stated explicitly?
- ✓ Is the 'flag, don't silently fix' data-quality principle stated explicitly?

11. Assumptions & Constraints

- The five source datasets are treated as already acquired to their documented specification; synthetic-data generation itself is out of scope of this PRD.
- No predictive or machine-learning component is included in v1 — the platform's purpose is insight generation and evidence-backed recommendation, not forecasting.
- All monetary, seasonal, and behavioural patterns are expected to reflect Indian gym-industry realities (pricing in INR, city-tier effects, festival/monsoon seasonality).
- CSV-only persistence is a non-negotiable constraint across the entire pipeline — no parquet, pickle, or other binary formats at any stage.
- The Streamlit app performs no analytical computation of its own; every KPI, segment, and flagged risk it displays was computed and validated upstream in the notebooks.